

PROJECT: BEE-PC

sub-system: EXP

Version: 1.0.0



Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 01_COVER PAGE.SchDoc	Sheet 1 of 28
Checked by: Bao Bui	Document No: 1	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

1	2	3	4	5	6
A					
B					
C					
D					
1	2	3	4	5	6



Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 02_BLOCK DIAGRAM.Sch	Sheet 2 of 28
Checked by: Bao Bui	Document No: 2	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

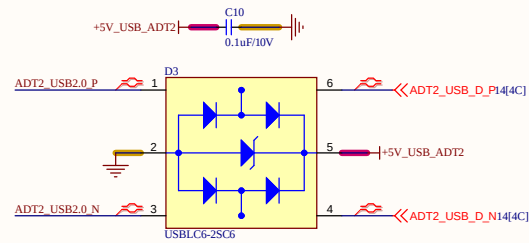
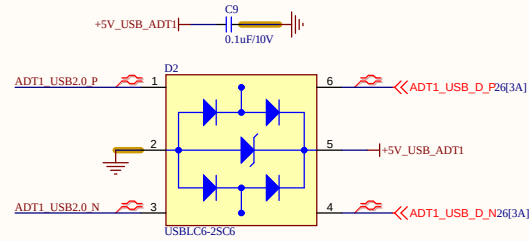
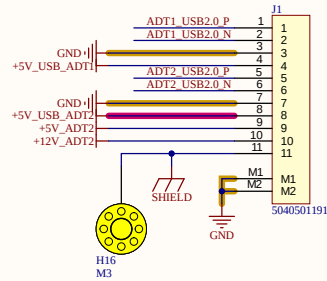
INDEX

Page	Title	Page	Title	Page	Title
1	COVER PAGE	11		21	
2	BLOCK DIAGRAM	12		22	
3	INDEX	13		23	
4	PI HAT	14		24	
5		15		25	
6		16		26	
7		17		27	
8		18		28	
9		19		29	
10		20		30	



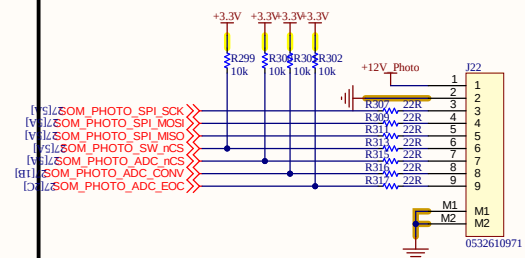
Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 03_INDEX.SchDoc	Sheet 3 of 28
Checked by: Bao Bui	Document No: 3	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

form Adaptor Board



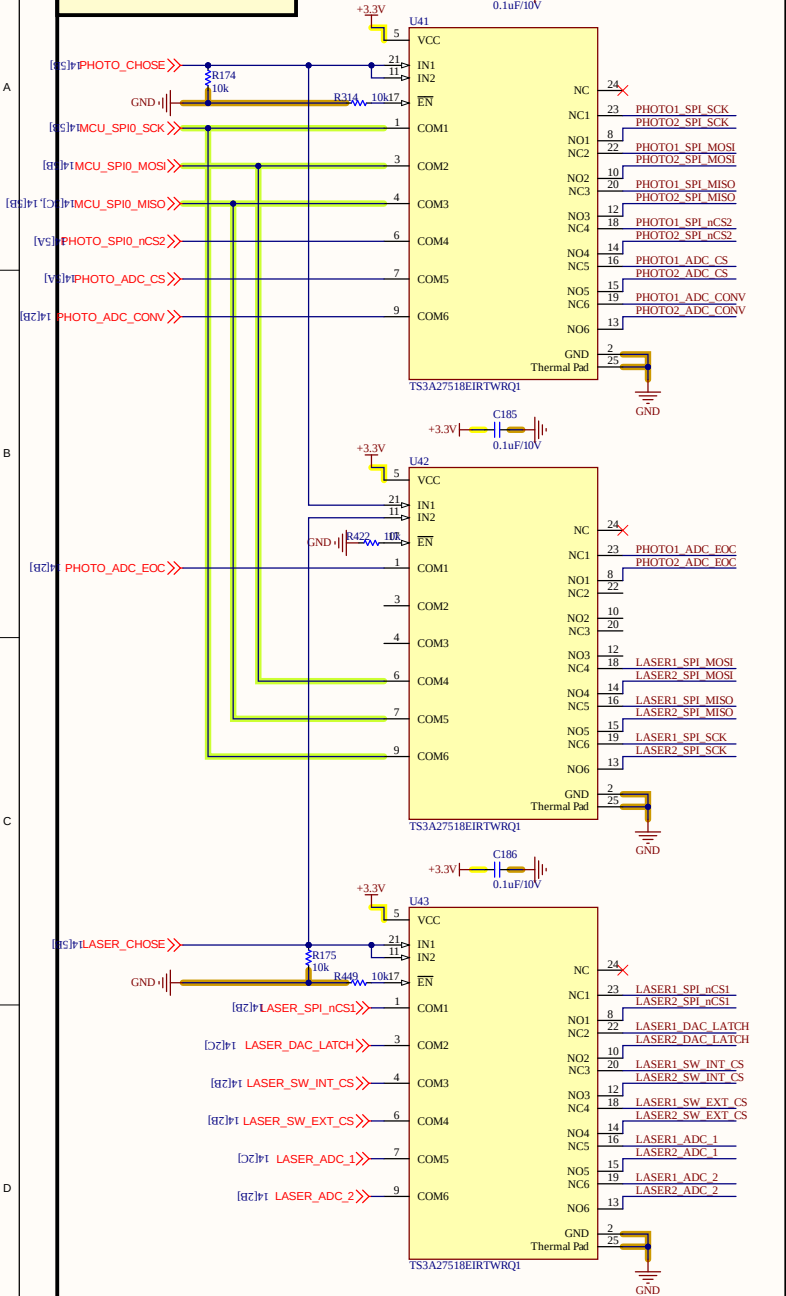
Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 04_CONNECTOR_IN.Sch	Sheet 4 of 28
Checked by: Bao Bui	Document No: 4	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

SOM to PHOTO

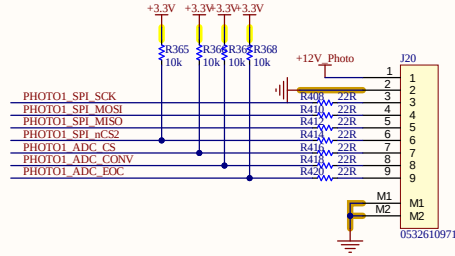


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 04_CONNECTOR_OUT.SchDoc	Sheet 5 of 28
Checked by: Bao Bui	Document No: 5	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

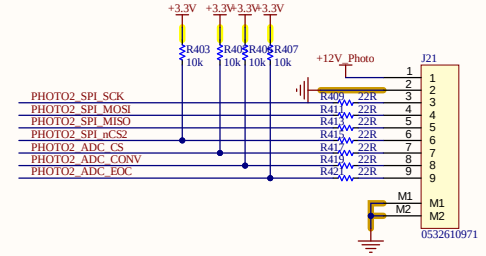
MCU SPI PHOTO



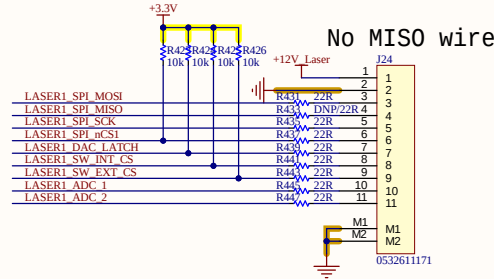
for PHOTO board



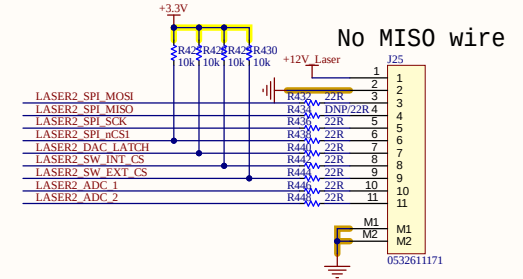
for PHOTO board



for LASER board



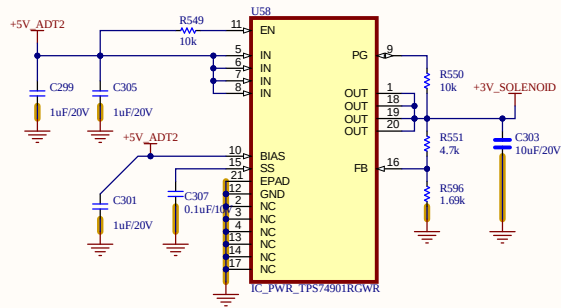
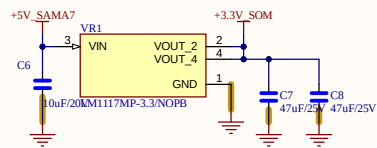
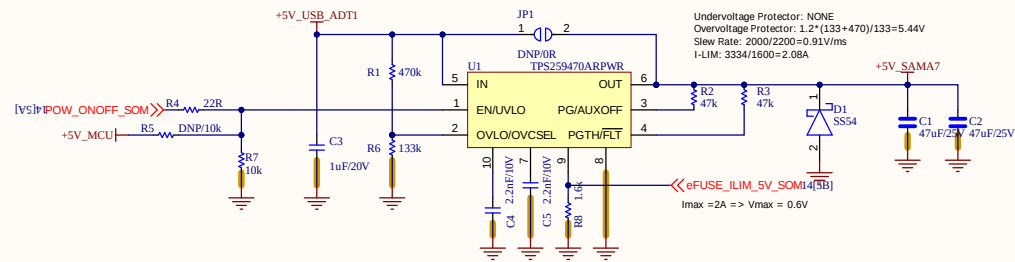
for LASER board



LASER_CHOSE 0: LASER1
1: LASER2
PHOTO_CHOSE 0: PHOTO1
1: PHOTO2



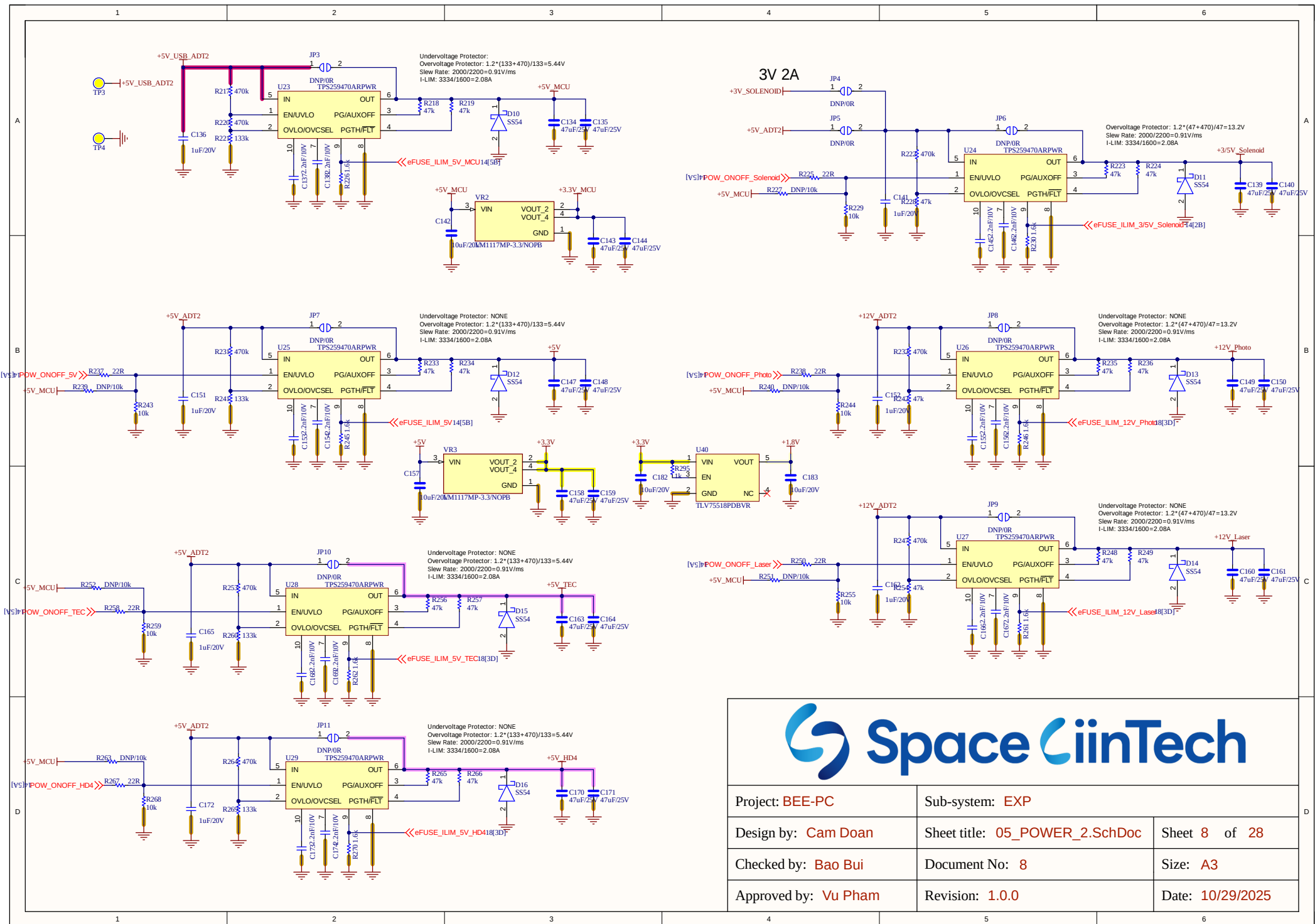
Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 04_CONNECTOR_OUT_25	Sheet No. of 28
Checked by: Bao Bui	Document No: 6	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025



NC	—	2, 3, 4, 13, 14, 17	—	—	No connection. This pin can be left floating or connected to GND to allow better thermal contact to the top-side plane.
----	---	---------------------	---	---	---

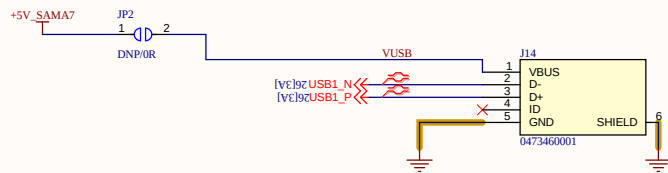


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 05_POWER.SchDoc	Sheet 7 of 28
Checked by: Bao Bui	Document No: 7	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

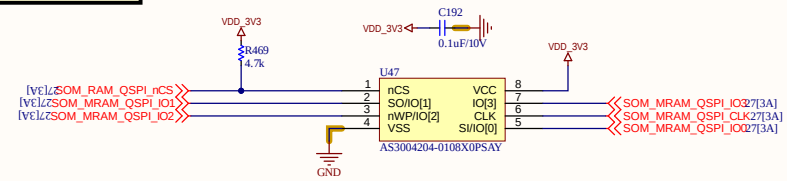


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 05_POWER_2.SchDoc	Sheet 8 of 28
Checked by: Bao Bui	Document No: 8	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

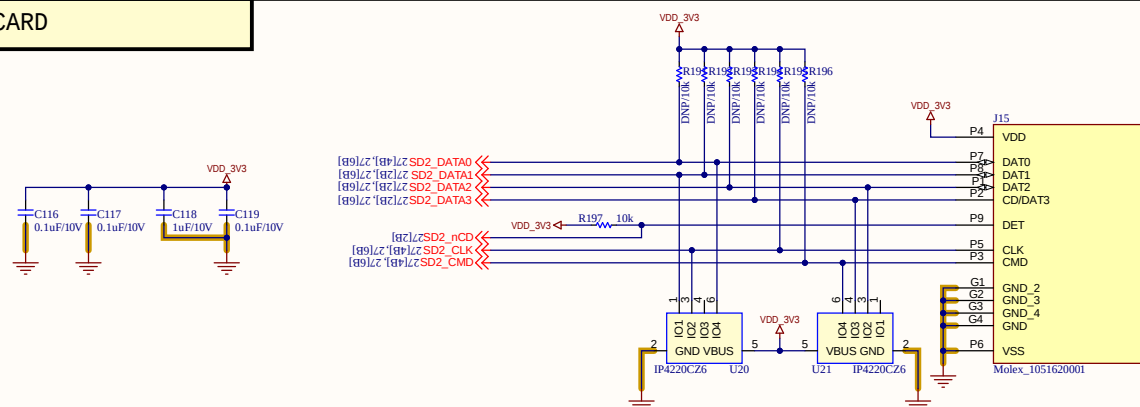
USB



MRAM 4Mbit



SD CARD



Project: BEE-PC

Sub-system: EXP

Design by: **Cam Doan**

Sheet title: 06_SOM_USB_CARD.Schema Sheet 9 of 28

Checked by: Bao Bui

Document No: 9

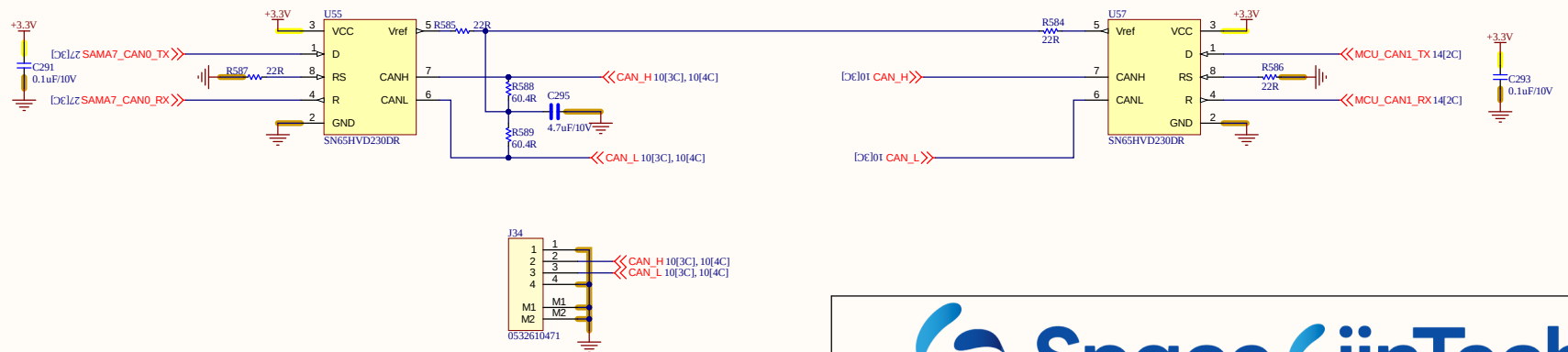
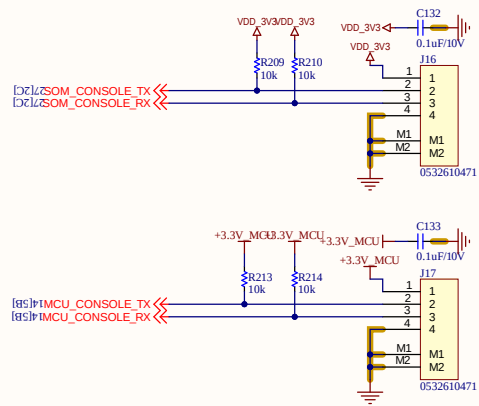
Size: A3

Approved by: **Vu Pham**

Revision: 1.0.0

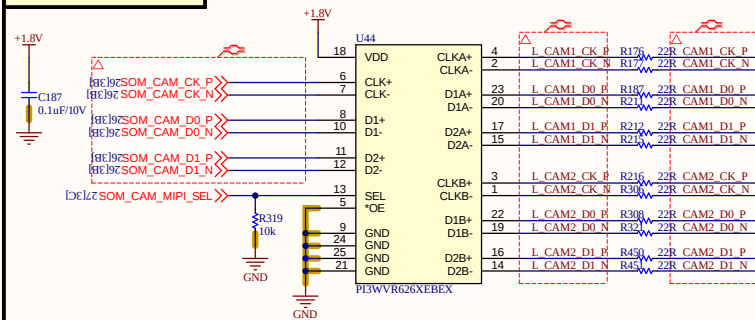
Date: 10/29/2025

CONSOLE

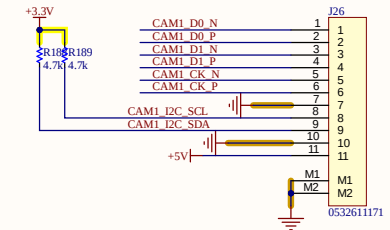


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 06_SOM_Console.SchDoc	Sheet 10 of 28
Checked by: Bao Bui	Document No: 10	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

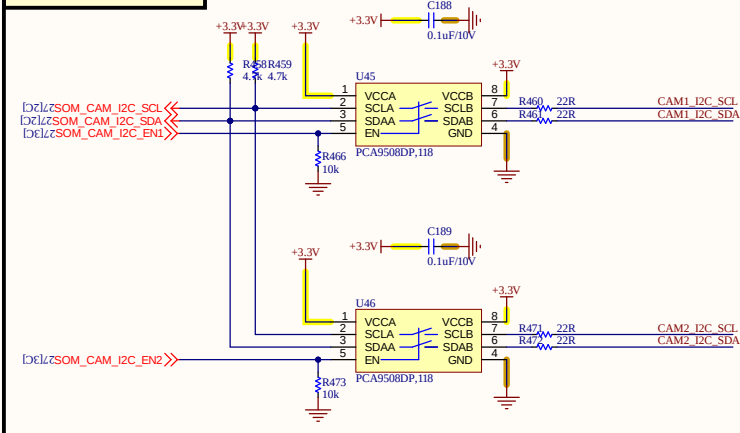
MIPI SWITCH



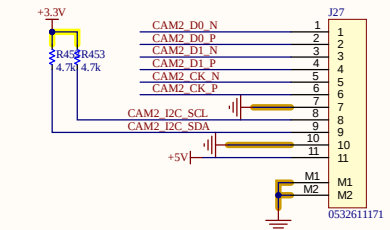
for CAMERA 1



I2C SWITCH

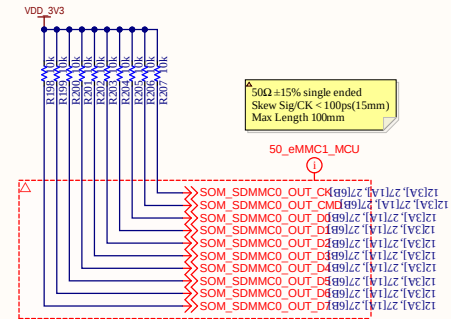
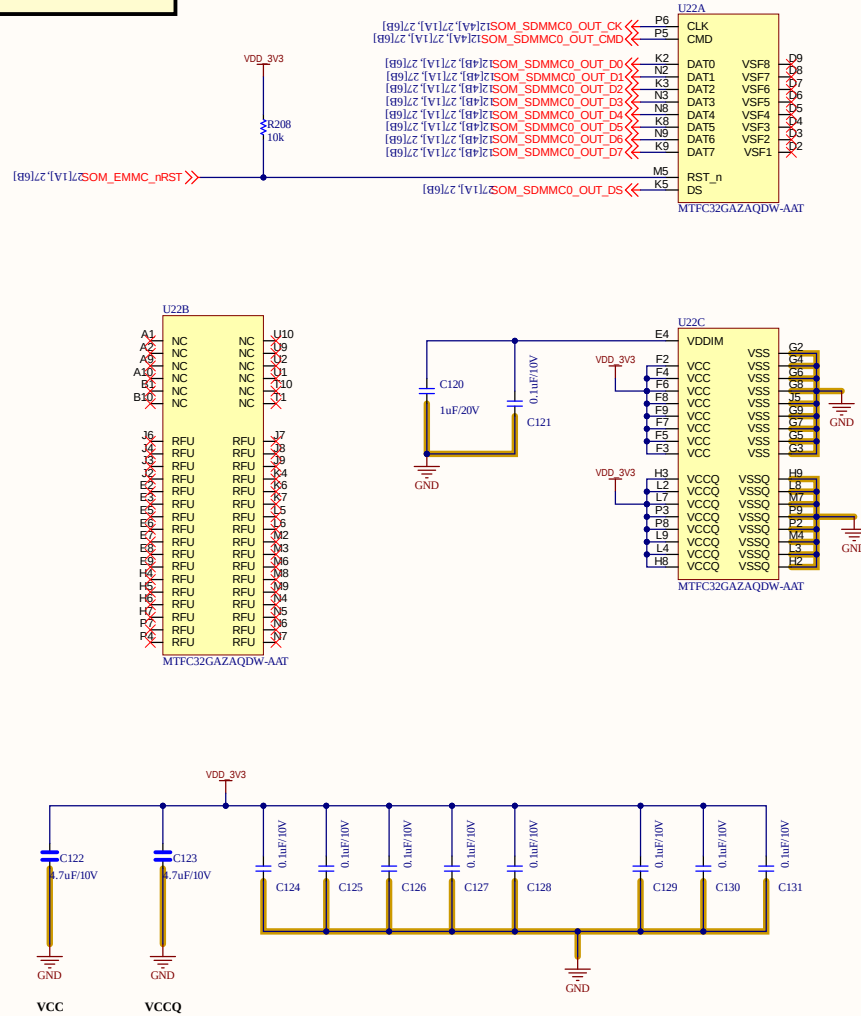


for CAMERA 2

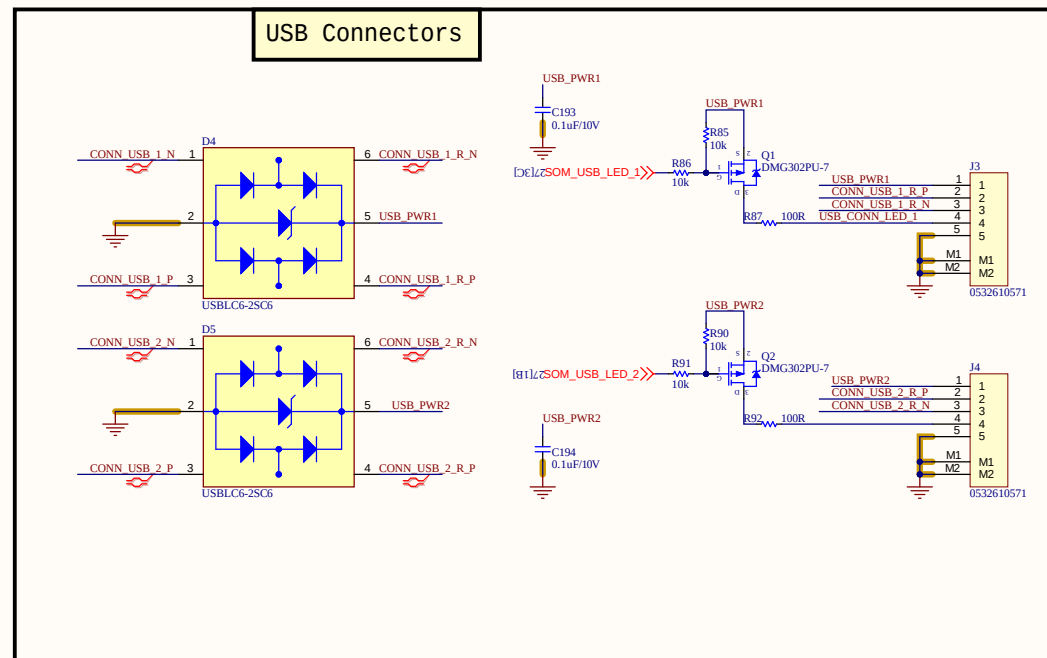
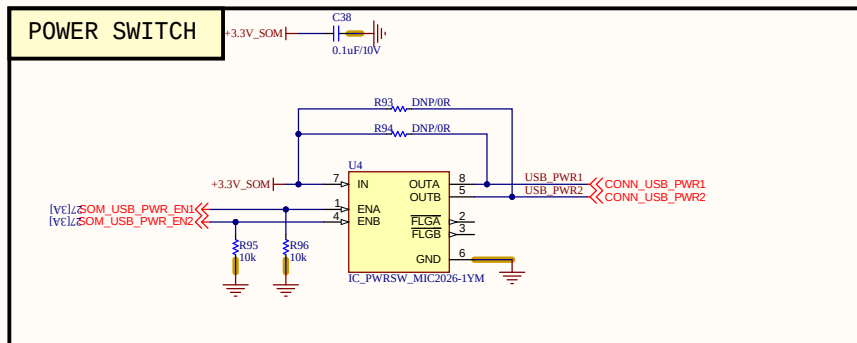
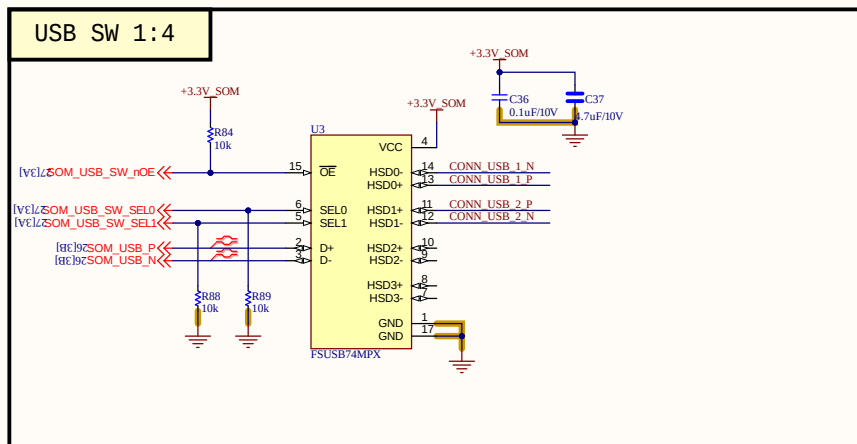


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 06_SOM_CAM.SchDoc	Sheet 11 of 28
Checked by: Bao Bui	Document No: 11	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

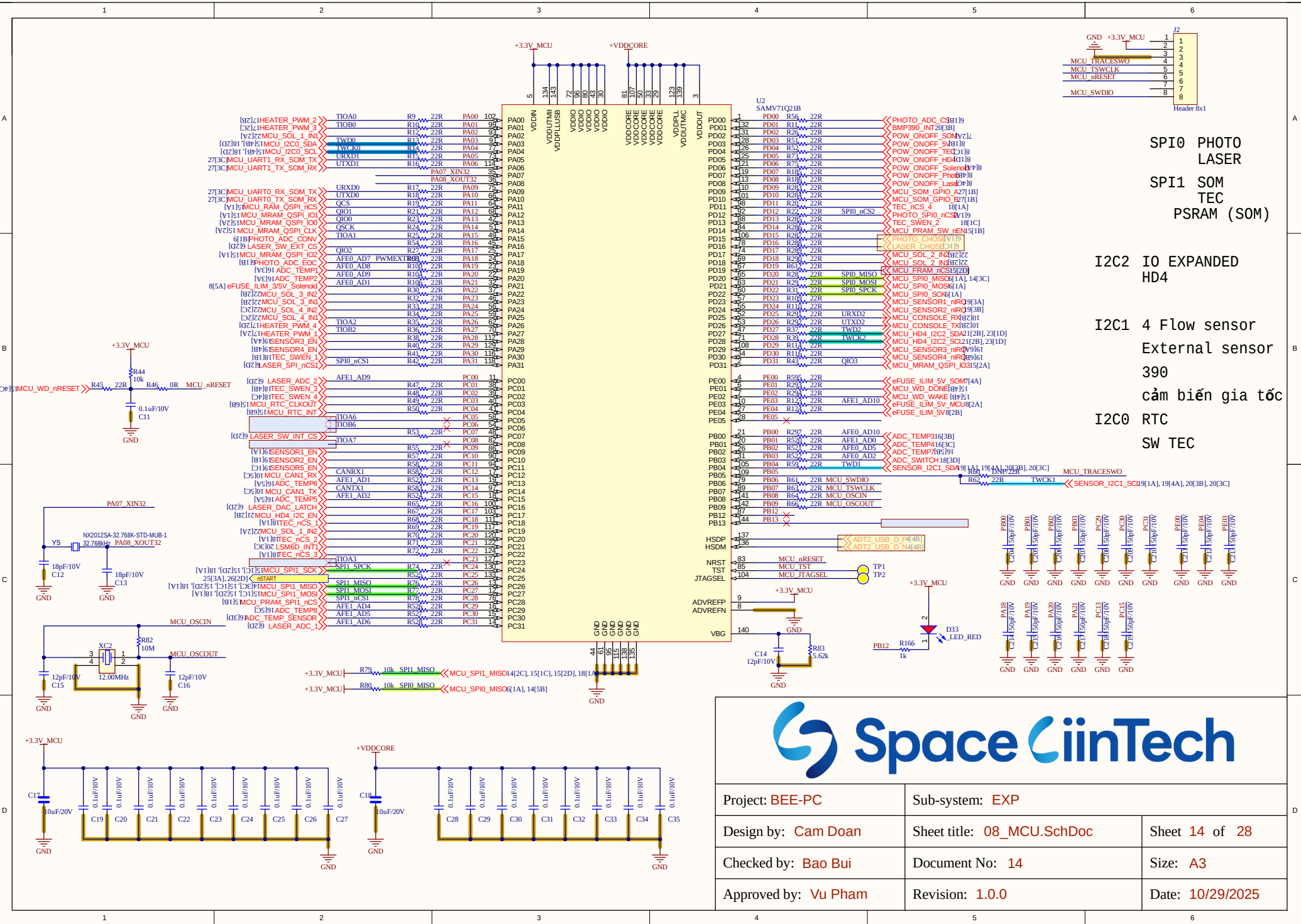
eMMC - Buffer



Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 07_eMMC.SchDoc	Sheet 12 of 28
Checked by: Bao Bui	Document No: 12	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025



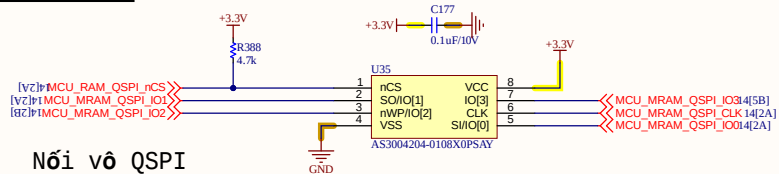
Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 07_USB.SchDoc	Sheet 13 of 28
Checked by: Bao Bui	Document No: 13	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025



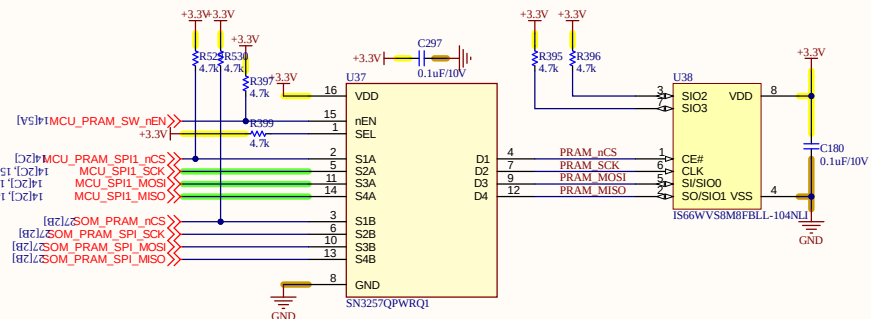
Project: BEE-PC		Sub-system: EXP	
Design by: Cam Doan		Sheet title: 08_MCU.SchDoc	
Checked by: Bao Bui		Document No: 14	
Approved by: Vu Pham		Revision: 1.0.0	
		Sheet 14 of 28	
		Size: A3	
		Date: 10/29/2025	

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
V _{DDCORE}	Voltage Range	Full operational	1.7		VDDIN	V

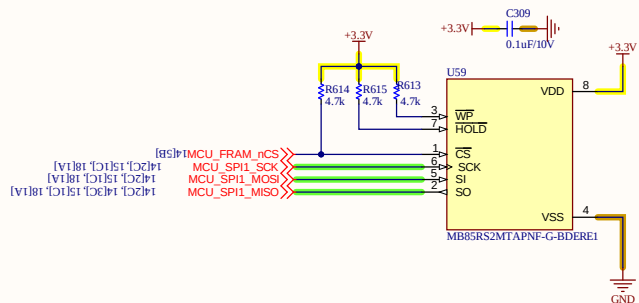
MRAM 4Mbit



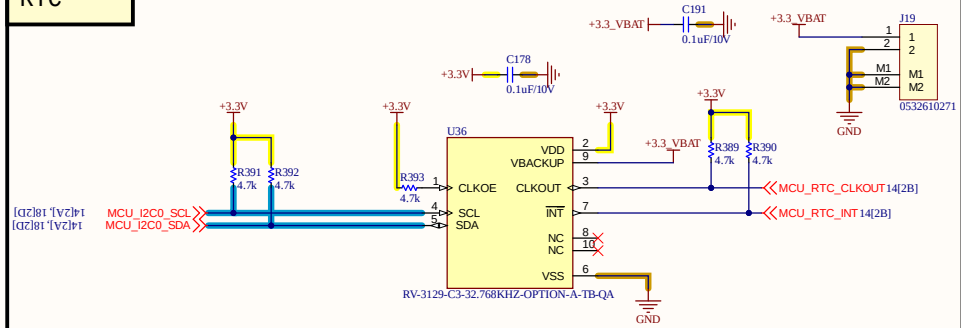
PRAM



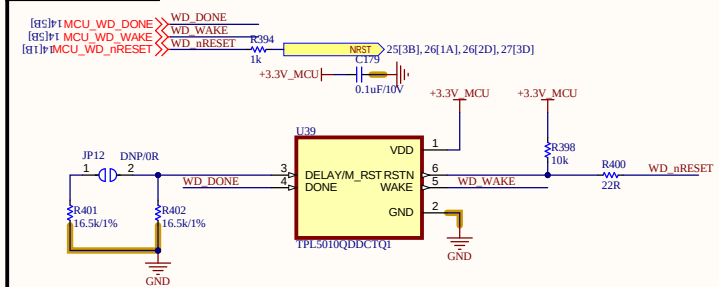
FRAM



RTC

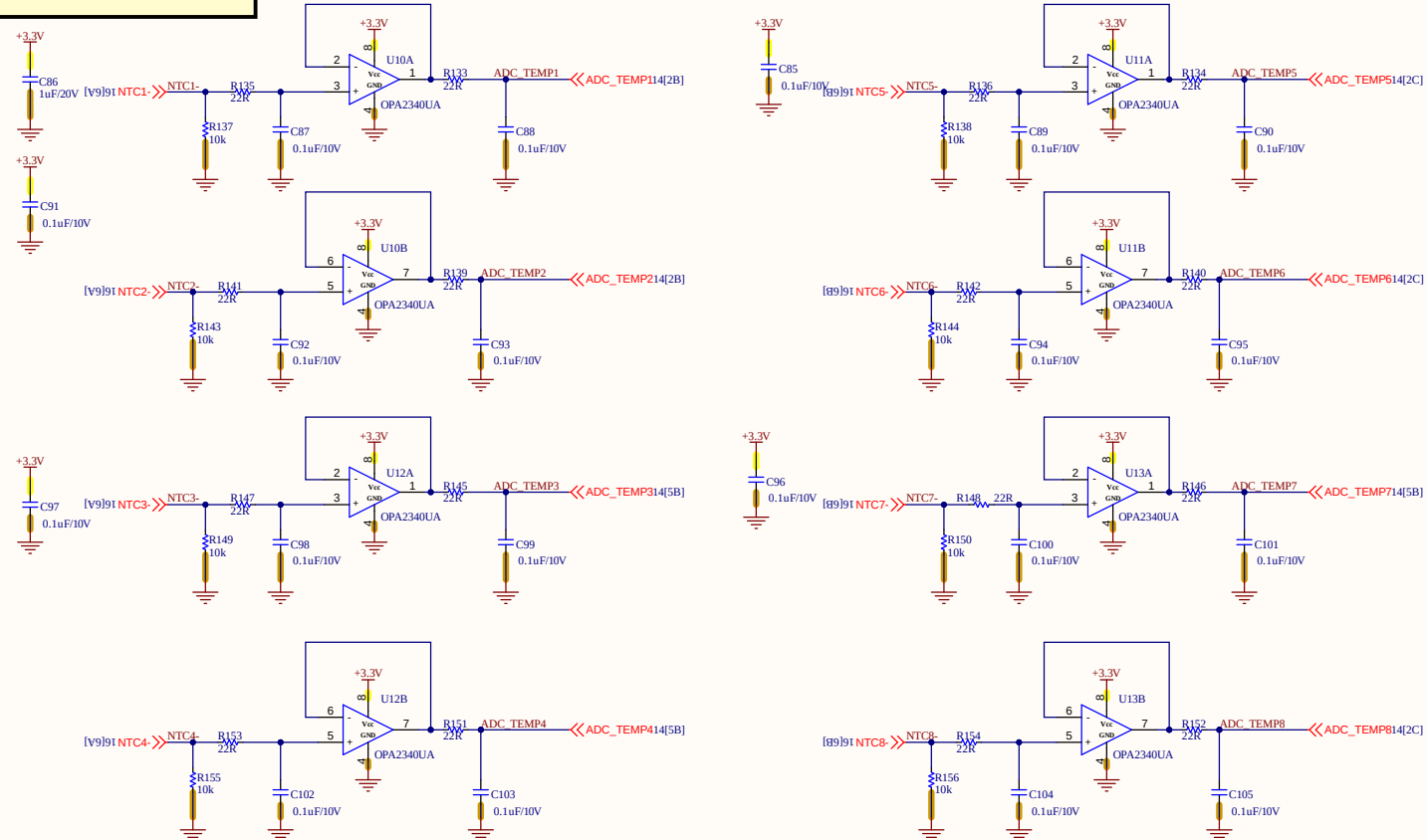


WATCH DOG

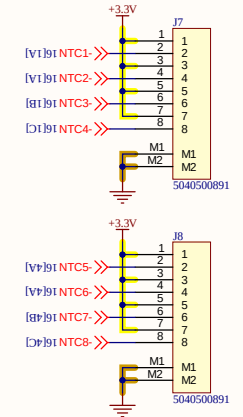


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 09_MRAM_RTC_WDT_Sch	Sheet 15 of 28
Checked by: Bao Bui	Document No: 15	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

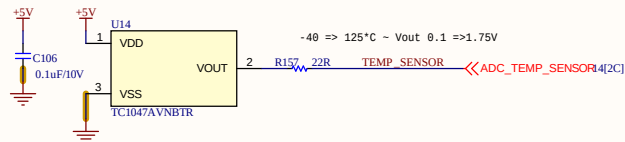
NTC



CONNECTOR FOR NTC

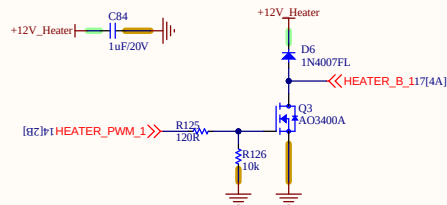


TEMPERATURE SENSOR

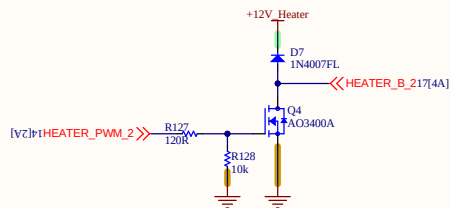


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 10_NTC_TEMPSENSOR	Sheet 16 of 28
Checked by: Bao Bui	Document No: 16	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

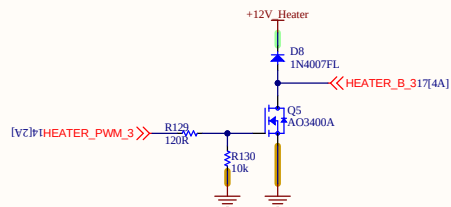
HEATER 1



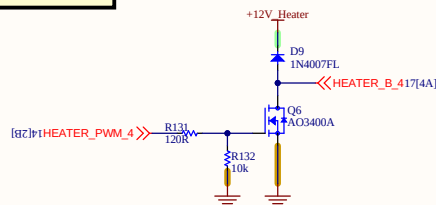
HEATER 2



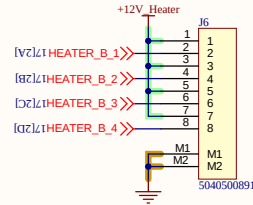
HEATER 3



HEATER 4



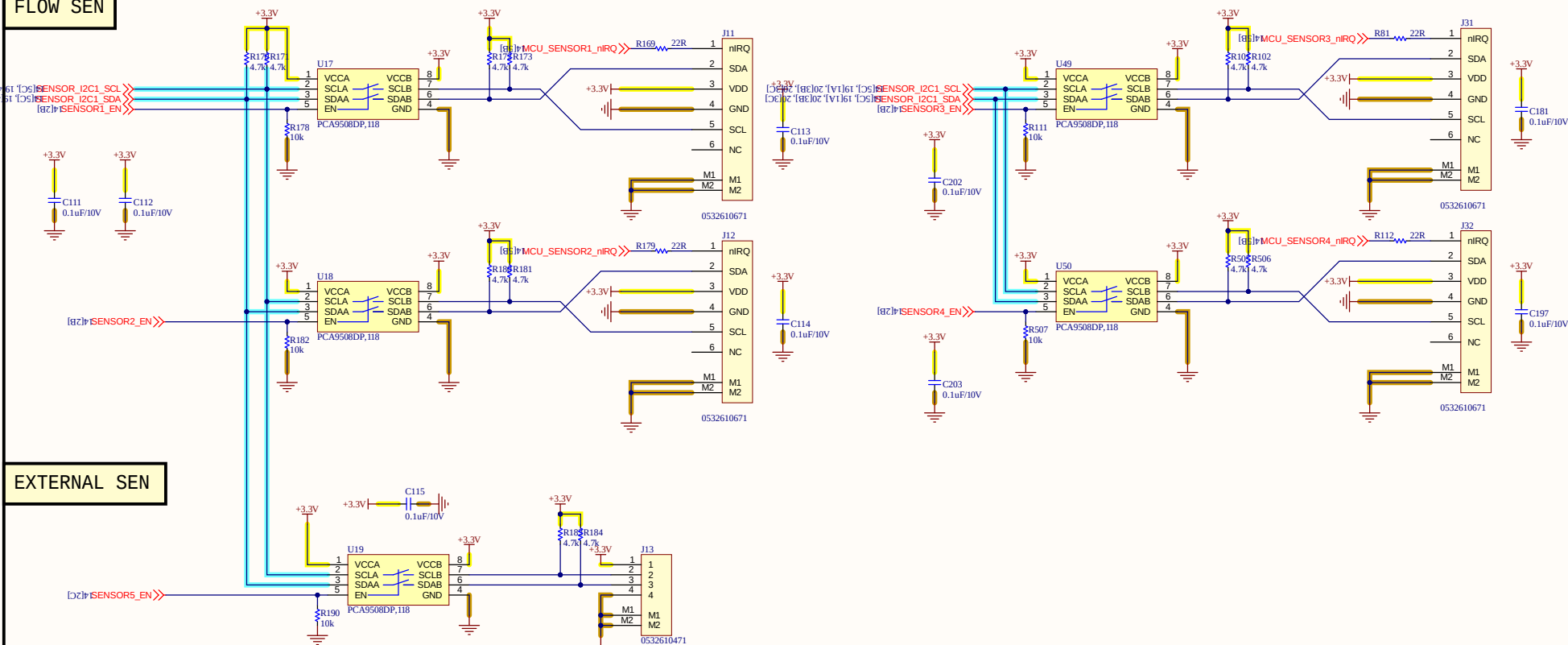
CONNECTOR FOR NTC



Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 11_HEATER.SchDoc	Sheet 17 of 28
Checked by: Bao Bui	Document No: 17	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

FLOW SEN

EXTERNAL SEN



Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 13_FLOW_Sensor.SchDoc	Sheet 19 of 28
Checked by: Bao Bui	Document No: 19	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

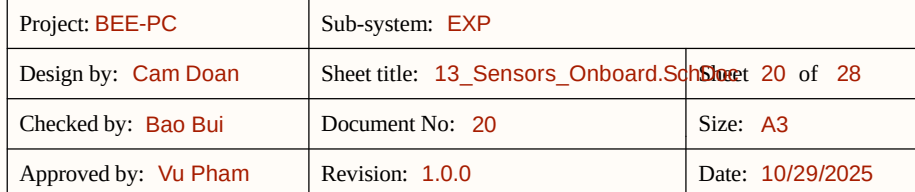
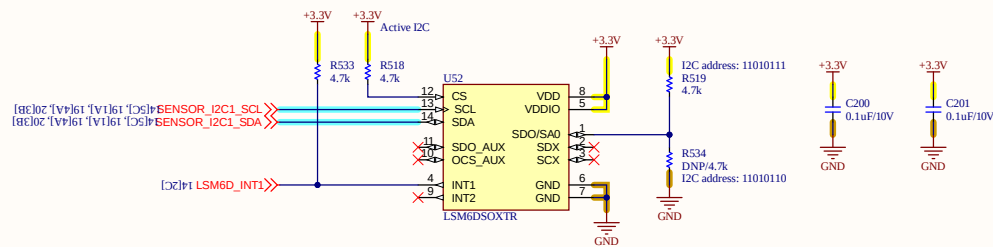
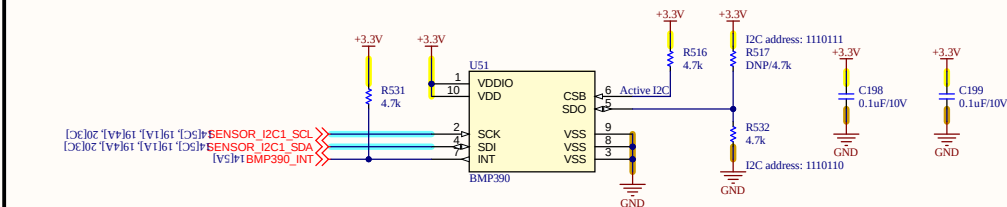
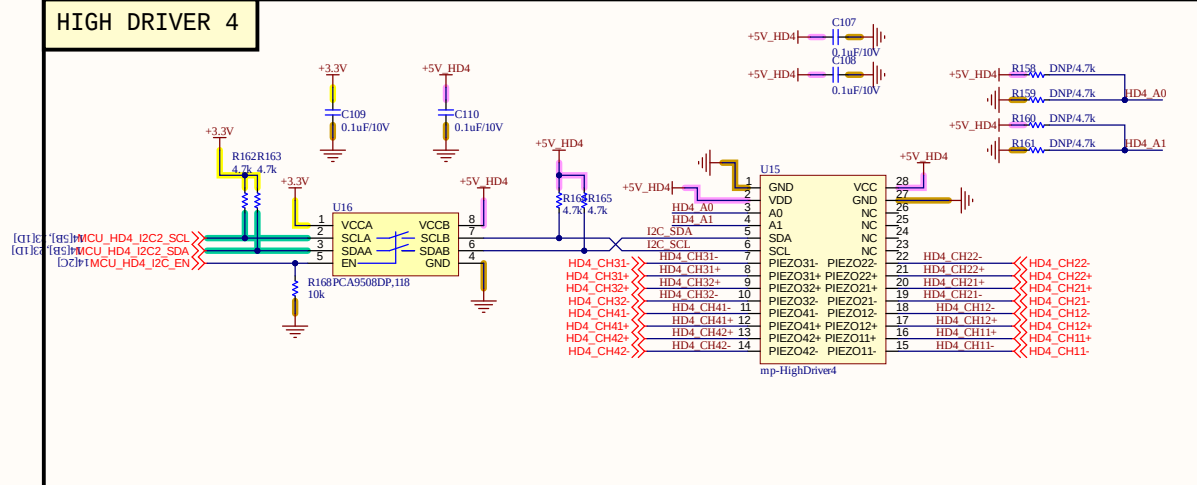
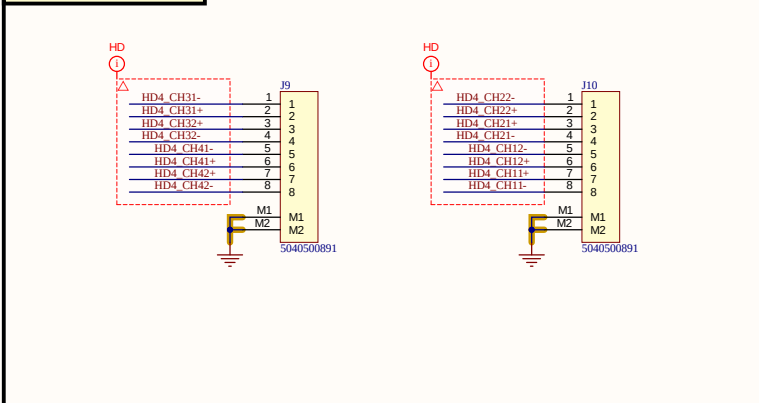


Figure 24. LSM6DSOX electrical connections in mode 1

HIGH DRIVER 4

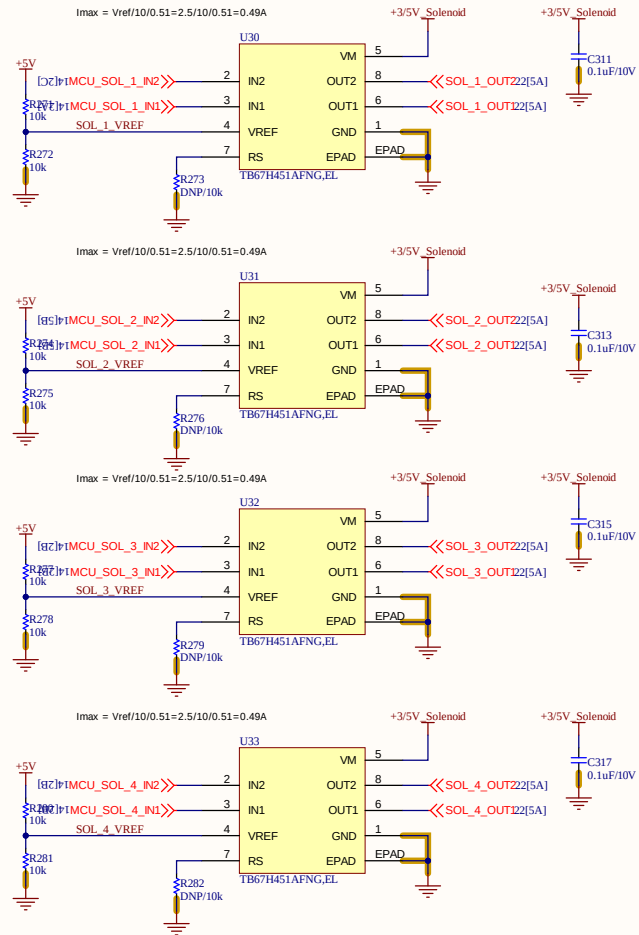


CONNECTOR BUMP

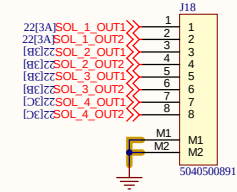


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 14_HD4_PUMP.SchDoc	Sheet 21 of 28
Checked by: Bao Bui	Document No: 21	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

SOLENOID



CONNECTOR SOLENOID

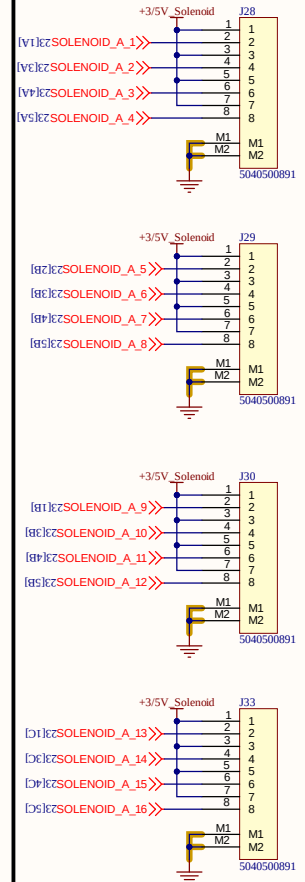


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 15_SOLEDNOID.SchDoc	Sheet 22 of 28
Checked by: Bao Bui	Document No: 22	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

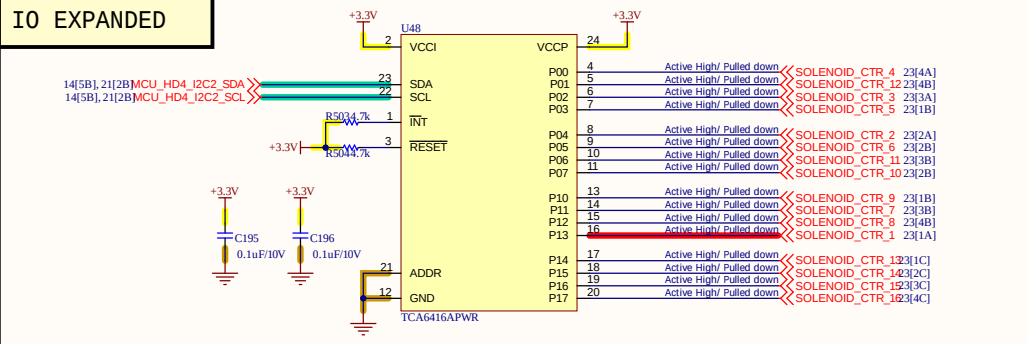
SOLENOID x16



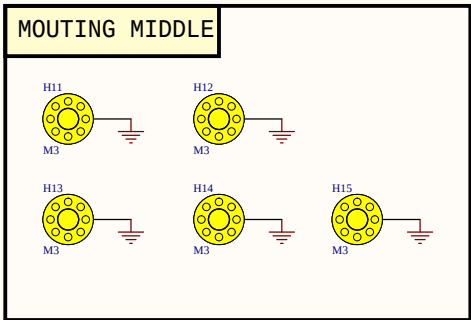
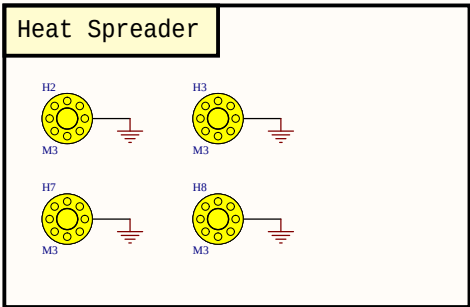
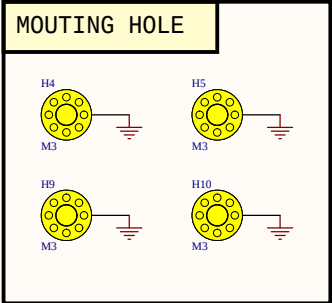
CONN SOLENOID



IO EXPANDED

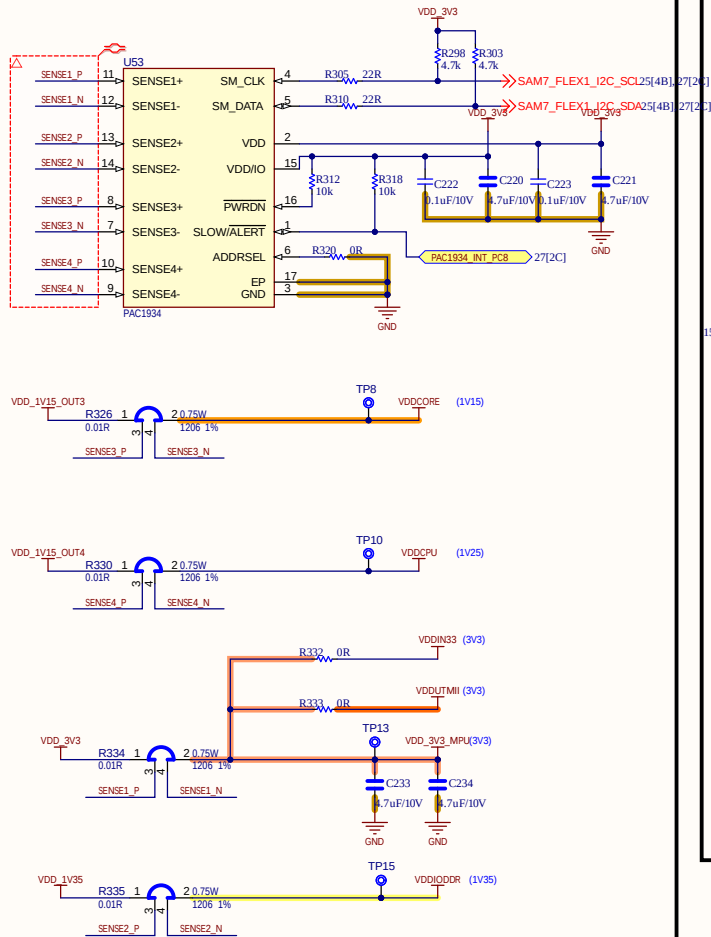


Project: BEE-PC	Sub-system: EXP
Design by: Cam Doan	Sheet title: 15_SOLENOID 2.SchDoc
Checked by: Bao Bui	Document No: 23
Approved by: Vu Pham	Revision: 1.0.0
	Size: A3
	Date: 10/29/2025

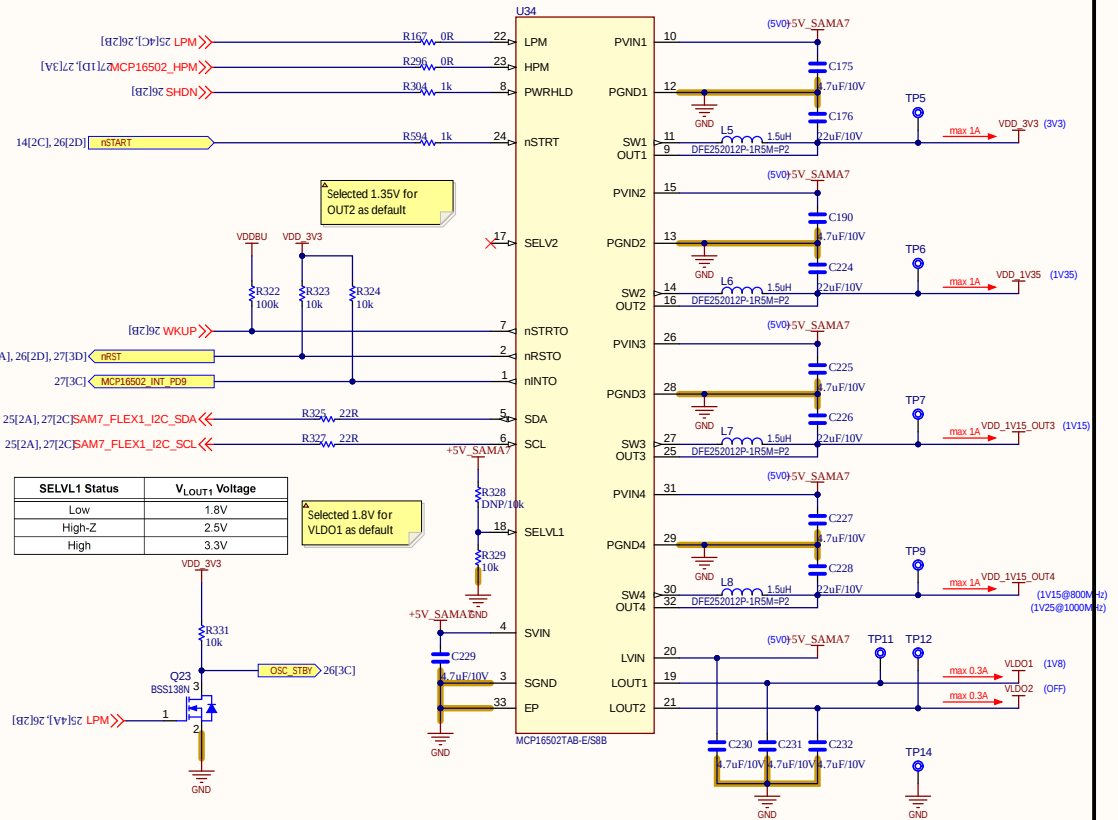


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: MOUTING.SchDoc	Sheet 24 of 28
Checked by: Bao Bui	Document No: 24	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

Power Measurement and Distribution



PMIC

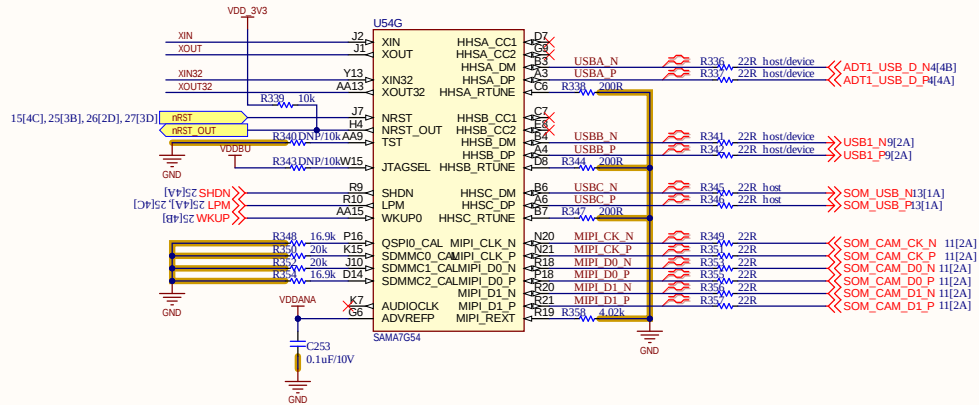


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 06_SAMA7G54-PMIC.Sch	Sheet 25 of 28
Checked by: Bao Bui	Document No: 25	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

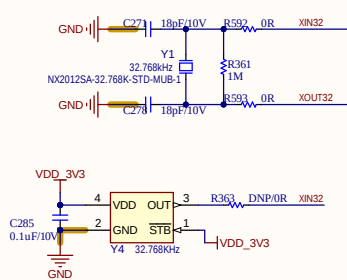
Thêm CAN giữa 2 khối

MPU configuration

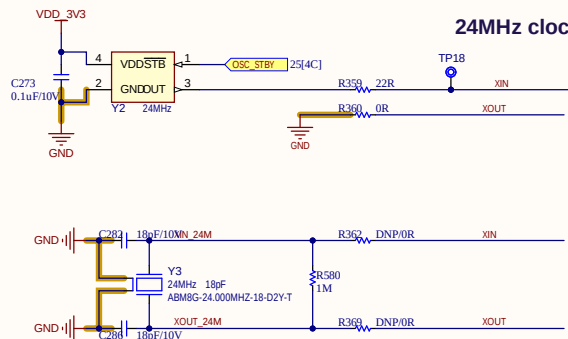
MPU decoupling



32.768KHz clock



24MHz clock

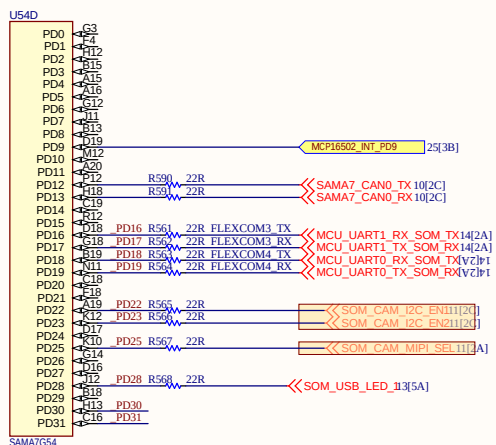
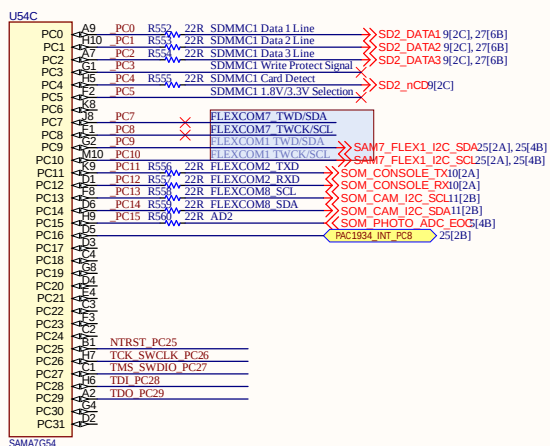
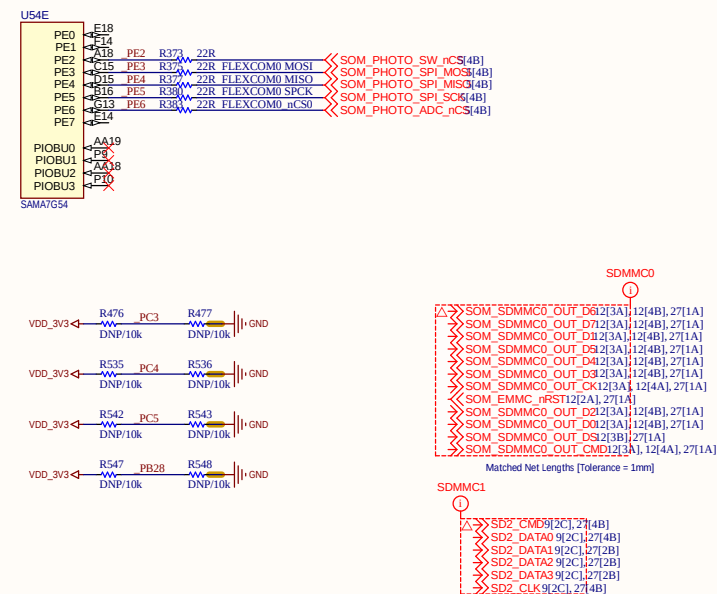
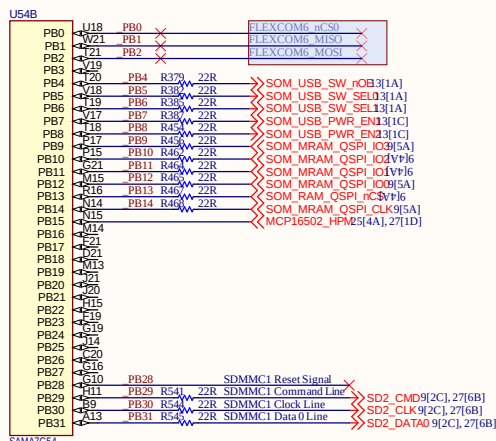


+3.3_VBAT



Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 06_SAMA7G54-MPU.Sch	Sheet 26 of 28
Checked by: Bao Bui	Document No: 26	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

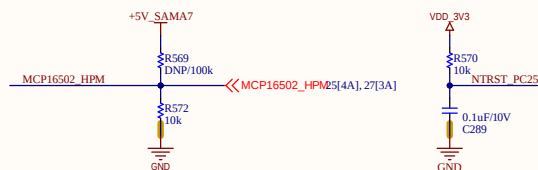
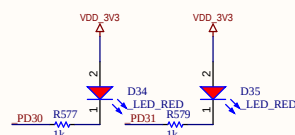
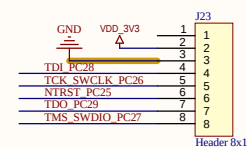
PA0	H17	PA0	R371	22R	>>> SOM_SDMCMCO_OUT_OU2[33A], 12[4A], 27[6B]
PA1	G20	PA1	R37	22R	>>> SOM_SDMCMCO_OUT_OU2[33A], 12[4A], 27[6B]
PA2	H13	PA2	R37	22R	>>> SOM_EMMC_nRST[21A], 27[6B]
PA3	H14	PA3	R37	22R	>>> SOM_SDMCMCO_OUT_OU2[33A], 12[4A], 27[6B]
PA4	H15	PA4	R37	22R	>>> SOM_SDMCMCO_OUT_OU2[33A], 12[4A], 27[6B]
PA5	G21	PA5	R37	22R	>>> SOM_SDMCMCO_OUT_OU2[33A], 12[4A], 27[6B]
PA6	H14	PA6	R37	22R	>>> SOM_SDMCMCO_OUT_OU2[33A], 12[4A], 27[6B]
PA7	H16	PA7	R37	22R	>>> SOM_SDMCMCO_OUT_OU2[33A], 12[4A], 27[6B]
PA8	G20	PA8	R47	22R	>>> SOM_SDMCMCO_OUT_OU2[33A], 12[4A], 27[6B]
PA9	H15	PA9	R47	22R	>>> SOM_SDMCMCO_OUT_OU2[33A], 12[4A], 27[6B]
PA10	G20	PA10	R47	22R	>>> SOM_SDMCMCO_OUT_OU2[33A], 12[4A], 27[6B]
PA11	G20	PA11	R47	22R	>>> SOM_SDMCMCO_OUT_OU2[33A], 12[4A], 27[6B]
PA12	V14				>>> SOM_SDMCMCO_OUT_DS[39B], 27[6B]
PA13	A20				
PA14	A20				
PA15	V15				
PA16	V16	PA17	R47	22R	TIOA1 <<< SOM_PHOTO_ADC_CONV[54B]
PA17	V16	PA18	R47	22R	<<< MCU_SOM_GPIO_B[5A]
PA18	V16				
PA19	G21	PA20	R47	22R	TIOA2 <<< SOM_USB_LED_3[5B]
PA20	G21				
PA21	V18	PA22	R47	22R	<<< MCU_SOM_GPIO_A[5A]
PA22	V18				
PA23	V19				
PA24	U04				
PA25	W16	PA26	R537	22R	FLEXCOM5_SPCK <<< SOM_PRAM_SPI_SCK[51C]
PA26	W18	PA27	R537	22R	FLEXCOM5_NPCS0 <<< SOM_PRAM_nCS[51C]
PA27	W18	PA28	R537	22R	FLEXCOM5_MOSI <<< SOM_PRAM_nCS[51C]
PA28	W19	PA29	R537	22R	FLEXCOM5_MISO <<< SOM_PRAM_SPI_MOSI[51C]
PA29	W14	PA30			FLEXCOM6_CS1 <<< SOM_PRAM_SPI_MISO[51C]
PA30	W20	PA31			FLEXCOM6_SPCK <<< X



SPI:3 F0: PE3456
F5: PA26-789
F6

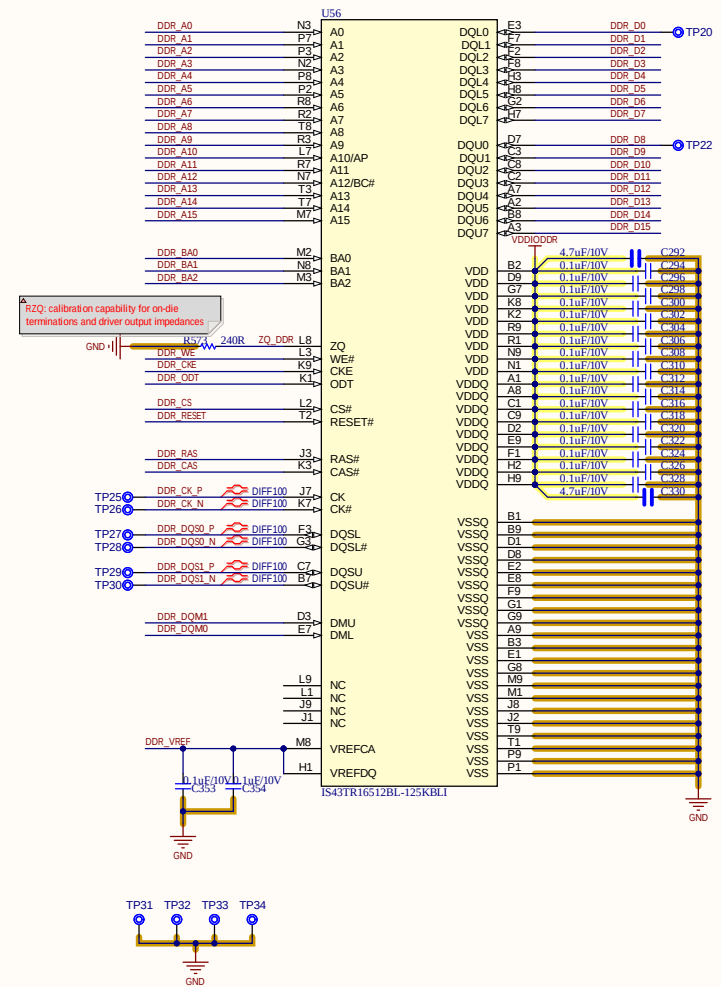
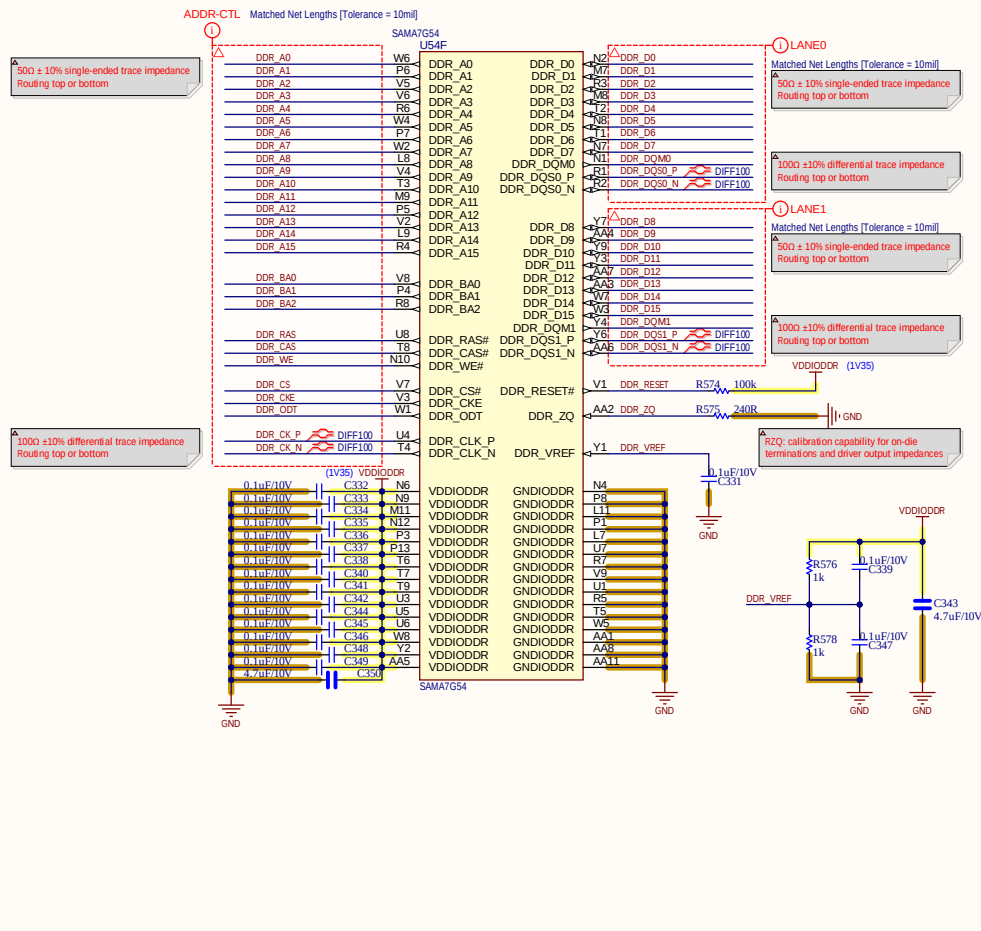
```
UART:3  F4:PD18 PD19
        F3:PD16 PD17
        F2:PC11 12
```

```
I2C:2    F1:PC9 10
          F8:PC1314
          F7:PC7 8
```



Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 06_SAMA7G54-PIO.SchDoc	Sheet 27 of 28
Checked by: Bao Bui	Document No: 27	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

SAMA7G55 DDR3-1066 controller



Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 06_SAMA7G54-DDR3.Sch	Sheet 28 of 28
Checked by: Bao Bui	Document No: 28	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

TP15 TP13 TP14 TP11 TP12 TP10 TP9 TP8 TP7 TP6 TP5

TP31 TP32 TP33 TP34

TP22 TP20

TP22
TP20
TP21
TP23
TP24
TP25
TP26
TP27
TP28

134

135

C317 C315 C313 C311

R612

R610 R608 R606 R604

U59
U58
U57
U56
U55
U54
U53
U52
U51
U50
U49
U48
U47
U46
U45
U44
U43
U42
U41
U40
U39
U38
U37
U36
U35
U34
U33
U32
U31
U30
U29
U28
U27
U26
U25
U24
U23
U22
U21
U20
U19
U18
U17
U16
U15
U14
U13
U12
U11
U10
U9
U8
U7
U6
U5
U4
U3
U2
U1

SpaceCiiTech
BEE-006 PAY-EXP V1.1.0

Designed by: Cam Doan
Checked by: Bao Bui
Approved by: Vu Pham

D504
D503
D502
D501
D500
D499
D498
D497
D496
D495
D494
D493
D492
D491
D490
D489
D488
D487
D486
D485
D484
D483
D482
D481
D480
D479
D478
D477
D476
D475
D474
D473
D472
D471
D470
D469
D468
D467
D466
D465
D464
D463
D462
D461
D460
D459
D458
D457
D456
D455
D454
D453
D452
D451
D450
D449
D448
D447
D446
D445
D444
D443
D442
D441
D440
D439
D438
D437
D436
D435
D434
D433
D432
D431
D430
D429
D428
D427
D426
D425
D424
D423
D422
D421
D420
D419
D418
D417
D416
D415
D414
D413
D412
D411
D410
D409
D408
D407
D406
D405
D404
D403
D402
D401
D400
D399
D398
D397
D396
D395
D394
D393
D392
D391
D390
D389
D388
D387
D386
D385
D384
D383
D382
D381
D380
D379
D378
D377
D376
D375
D374
D373
D372
D371
D370
D369
D368
D367
D366
D365
D364
D363
D362
D361
D360
D359
D358
D357
D356
D355
D354
D353
D352
D351
D350
D349
D348
D347
D346
D345
D344
D343
D342
D341
D340
D339
D338
D337
D336
D335
D334
D333
D332
D331
D330
D329
D328
D327
D326
D325
D324
D323
D322
D321
D320
D319
D318
D317
D316
D315
D314
D313
D312
D311
D310
D309
D308
D307
D306
D305
D304
D303
D302
D301
D300
D299
D298
D297
D296
D295
D294
D293
D292
D291
D290
D289
D288
D287
D286
D285
D284
D283
D282
D281
D280
D279
D278
D277
D276
D275
D274
D273
D272
D271
D270
D269
D268
D267
D266
D265
D264
D263
D262
D261
D260
D259
D258
D257
D256
D255
D254
D253
D252
D251
D250
D249
D248
D247
D246
D245
D244
D243
D242
D241
D240
D239
D238
D237
D236
D235
D234
D233
D232
D231
D230
D229
D228
D227
D226
D225
D224
D223
D222
D221
D220
D219
D218
D217
D216
D215
D214
D213
D212
D211
D210
D209
D208
D207
D206
D205
D204
D203
D202
D201
D200
D199
D198
D197
D196
D195
D194
D193
D192
D191
D190
D189
D188
D187
D186
D185
D184
D183
D182
D181
D180
D179
D178
D177
D176
D175
D174
D173
D172
D171
D170
D169
D168
D167
D166
D165
D164
D163
D162
D161
D160
D159
D158
D157
D156
D155
D154
D153
D152
D151
D150
D149
D148
D147
D146
D145
D144
D143
D142
D141
D140
D139
D138
D137
D136
D135
D134
D133
D132
D131
D130
D129
D128
D127
D126
D125
D124
D123
D122
D121
D120
D119
D118
D117
D116
D115
D114
D113
D112
D111
D110
D109
D108
D107
D106
D105
D104
D103
D102
D101
D100
D99
D98
D97
D96
D95
D94
D93
D92
D91
D90
D89
D88
D87
D86
D85
D84
D83
D82
D81
D80
D79
D78
D77
D76
D75
D74
D73
D72
D71
D70
D69
D68
D67
D66
D65
D64
D63
D62
D61
D60
D59
D58
D57
D56
D55
D54
D53
D52
D51
D50
D49
D48
D47
D46
D45
D44
D43
D42
D41
D40
D39
D38
D37
D36
D35
D34
D33
D32
D31
D30
D29
D28
D27
D26
D25
D24
D23
D22
D21
D20
D19
D18
D17
D16
D15
D14
D13
D12
D11
D10
D9
D8
D7
D6
D5
D4
D3
D2
D1

Board Stack Report